

genaut — frontier report

source CSV : genaut/logs/1state2ap2acc/default.csv
corpus rows: 83

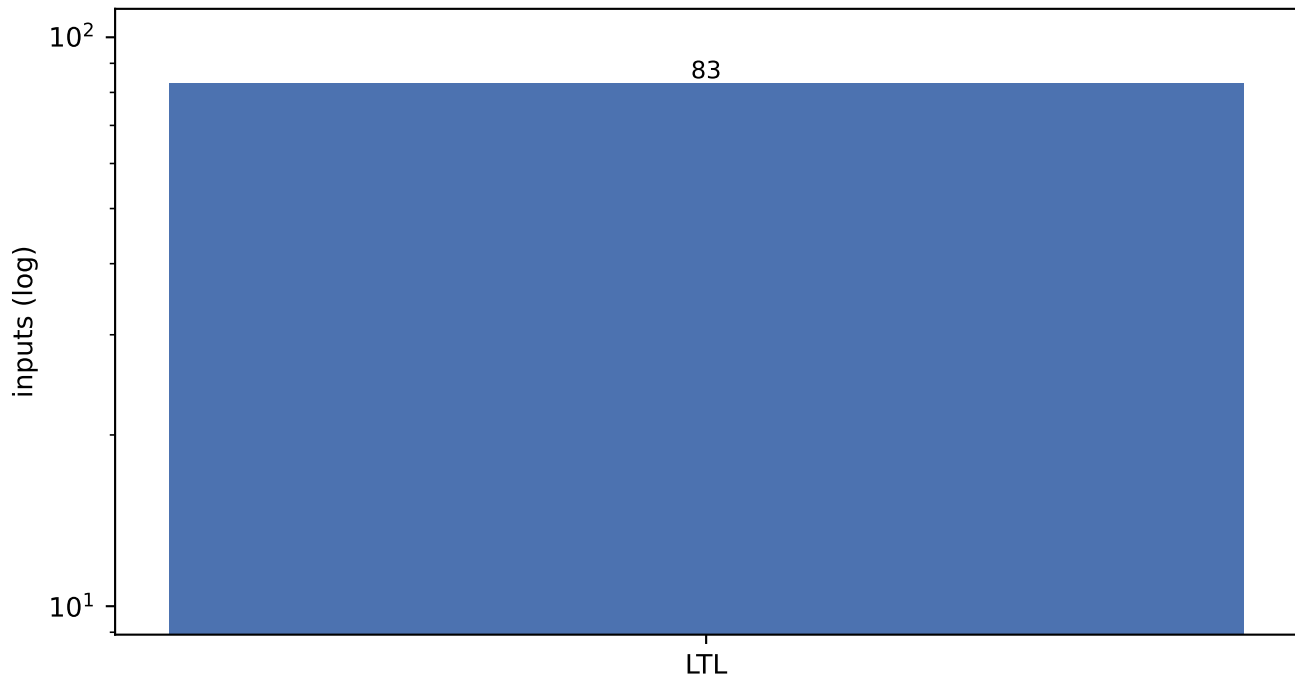
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 83
  LTL-definable (formula built) :    83  (100.0%)
    not-LTL (decided)           :      0  (  0.0%)
    ambiguous (timeout, unknown) :      0  (  0.0%)
decided (LTL or not-LTL): 83  (100.0%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 83 (0 with a gated/unprintable string)

distinct structures (md5): 76

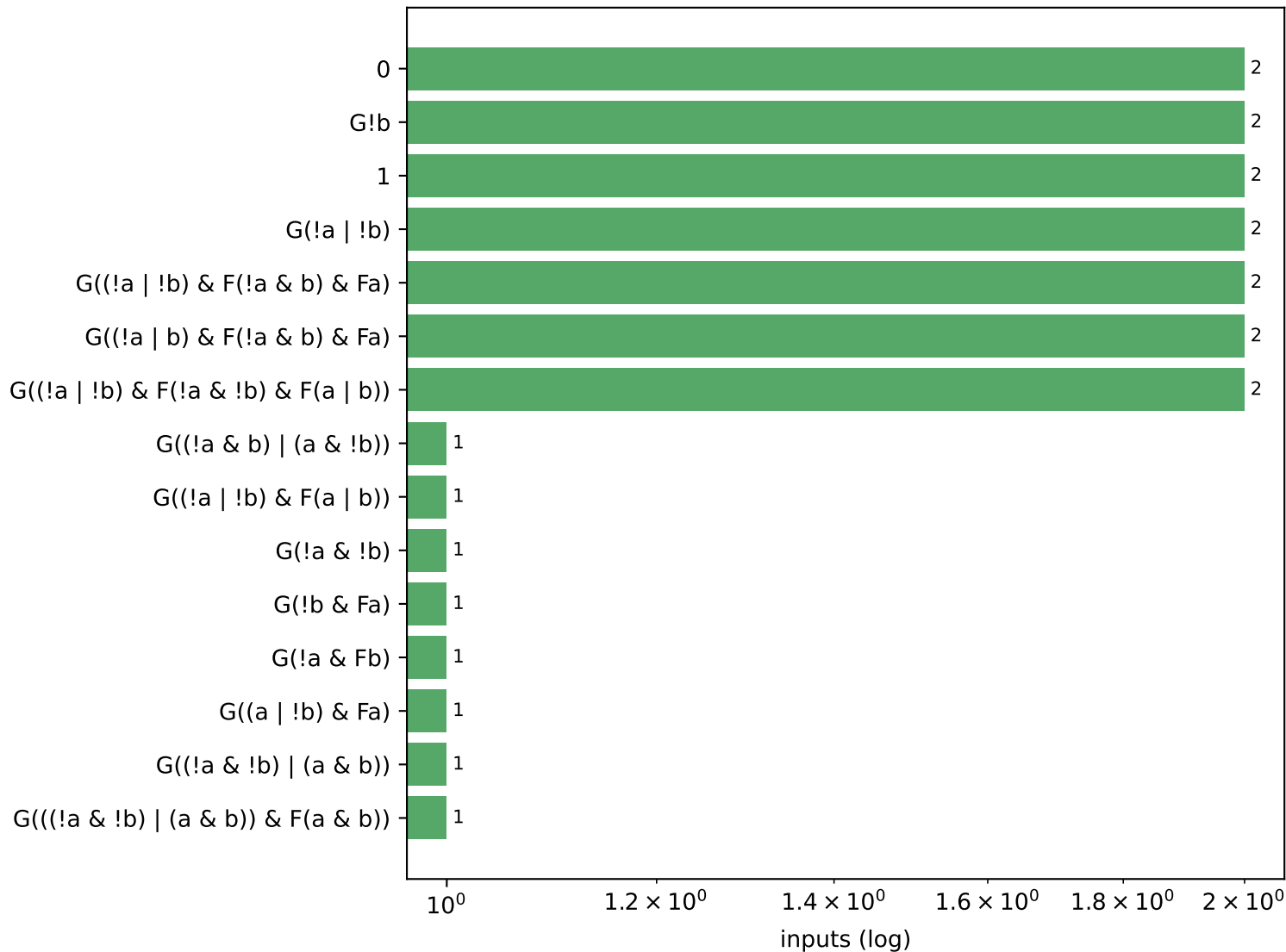
80% of answers covered by the top 60 structure(s)

90% of answers covered by the top 68 structure(s)

top 15 structures:

```
2 ( 2.4%) 0
2 ( 2.4%) G!b
2 ( 2.4%) 1
2 ( 2.4%) G(!a | !b)
2 ( 2.4%) G(!a | !b) & F(!a & b) & Fa)
2 ( 2.4%) G(!a | b) & F(!a & b) & Fa)
2 ( 2.4%) G(!a | !b) & F(!a & !b) & F(a | b))
1 ( 1.2%) G(!a & b) | (a & !b))
1 ( 1.2%) G(!a | !b) & F(a | b))
1 ( 1.2%) G(!a & !b)
1 ( 1.2%) G(!b & Fa)
1 ( 1.2%) G(!a & Fb)
1 ( 1.2%) G((a | !b) & Fa)
1 ( 1.2%) G(!a & !b) | (a & b))
1 ( 1.2%) G((!a & !b) | (a & b)) & F(a & b))
```

Top 15 reconstructed structures



Size frontier — flattened formula size

LTL answers with a size: 83

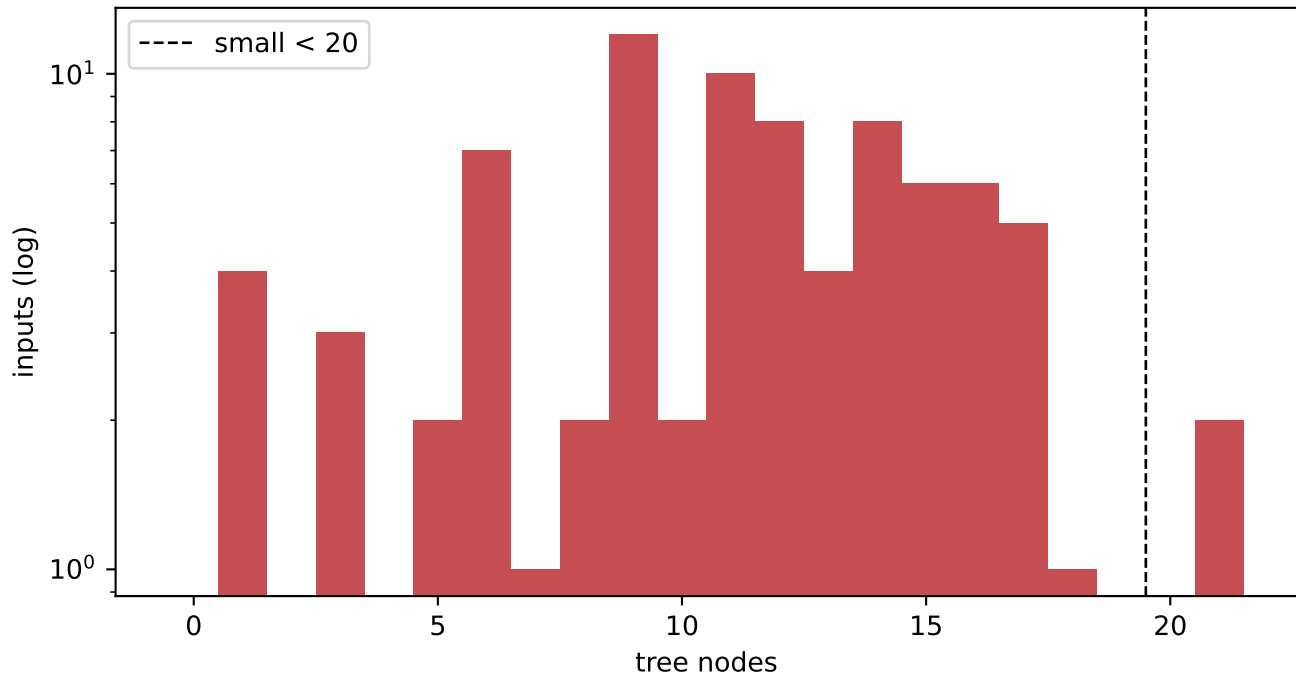
trivial (tree≤1, i.e. true/false): 4 (4.8%)

small (tree<20): 81 (97.6%)

larger (tree≥20): 2 (2.4%)

tree nodes: median 11 mean 11.1 max 22

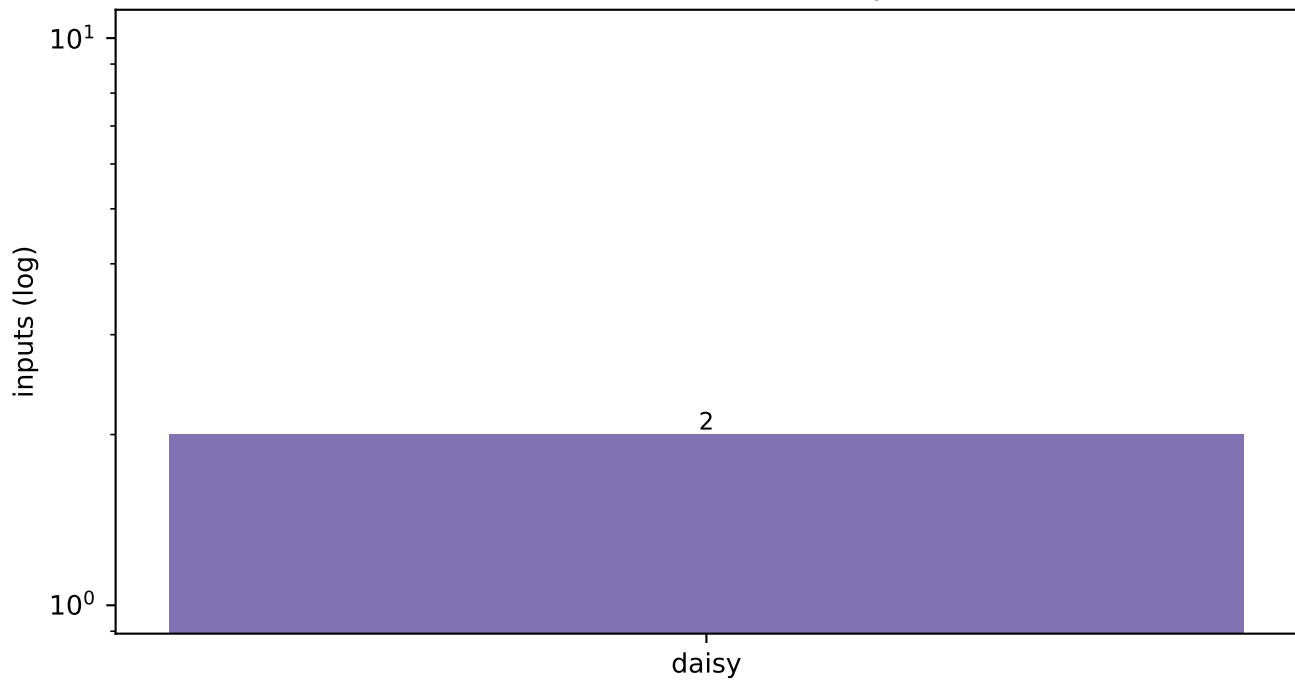
Flattened size (tree nodes); clipped at 21 (1 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 2  
  2 (100.0%) daisy  
build_s over true rows: total 0.1s  mean 0.075s  max 0.077s
```

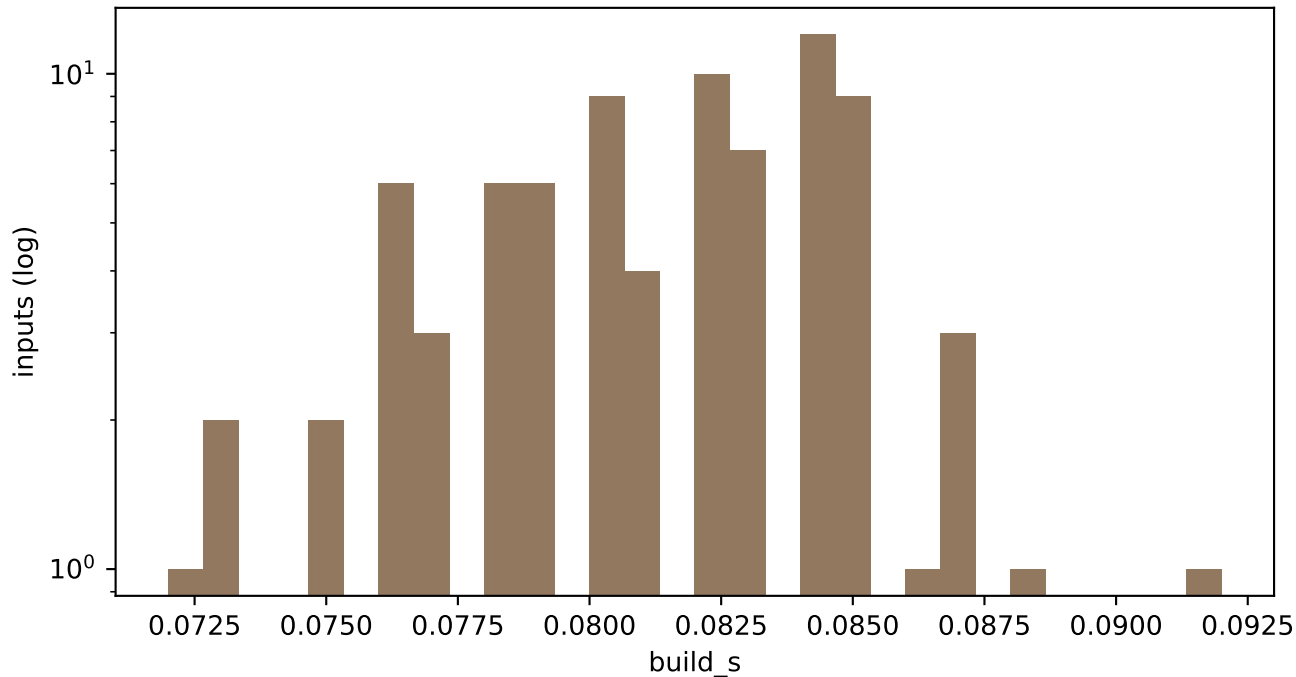

Routes to `true` (technique)



Build time & timeouts

```
total build: 6.7s    mean 0.081s    max 0.092s
  LTL          n= 83    mean 0.081s    max 0.092s
timeouts: 0
```

Build-time distribution



Verification — spot-oracle validation

validation over 83 LTL answers:
True 83 (100.0%)
NOT-equivalent (FALSE): 0 CLEAN